

Visual Analytics solution for Youtuber influencer marketing: Youtuber Finder

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1 Problem and idea

1.1 Problem

Nowadays companies use different strategies to advertise their product, the ad market shifted aggressively in the past decades from television to the internet. In the first phases of www blogs and websites were the main source of ad traffic, but since the last few years one of the main strategy to promote a product or a service is through "Influencer marketing".

Influencer marketing provides an easier way of advertising product since the agreement is between an individual and a company and not two companies, there can be single ads or a sponsorship that lasts for a long time.

There is a problem though in this approach, the product or service might be directed to a specific niche or age range or maybe just a specific country. Also, when those requirements are met, the decision about what youtuber is most correct might still not be that easy: a youtuber with a lot of subscribers but few views might be a flag of an inactive channel, a channel with a lot of views but few comments might be a flag of low user retention, and so on...

How to find the right influencer that can best promulgate the product or service?

1.2 Idea

In recent years, YouTube has emerged as one of the most influential platforms for content creators and marketers alike. With millions of channels and billions of users, understanding the dynamics of YouTube activity has become increasingly important.

Many platform propose a way of finding youtubers by applying some filters on a database (more on that in section 4), we thought of a visual ana-

lytics solution that could provide more interactivity and understandability than the common solutions.

From the problem description is quite obvious that some info on the number of average likes, followers, comments and views are required, also some filters on country and main topic of the channel.

The proposed visual analytics solution aims to leverage data visualization techniques to provide a holistic view of YouTube channel activity. By aggregating and visualizing metrics such as likes, comments, followers, and views, the solution enables users to identify outliers, uncover trends, and make data-driven decisions. More details in section 2

2 Solution

2.1 General description

The user can decide what dataset to use between a set of 4 different datasets for 4 different months. The solution provides 2 checkbox list filters (one for country and one for main category of videos made by the channel). Then there are the main visualization that are 5: one scatter plot where the youtubers are shown using MDS dimensionality reduction technique (the mds is done by default on 4 fields, but the user can choose to keep only some, down to 1); the other 4 visualization are histograms where each histogram represents the number of youtubers that are in some range of likes, views, comments and followers (the scale for each histogram can be changed between linear and logarithmic). The user can brush over the scatter plot and the histograms to reduce or increment the selection of youtubers he wants to look after.

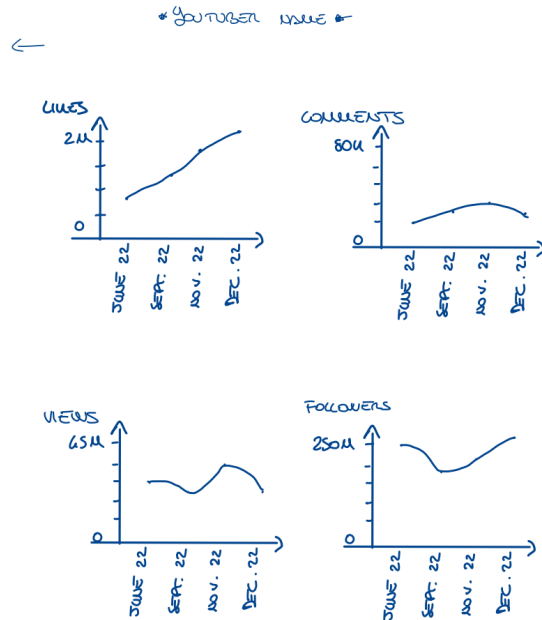
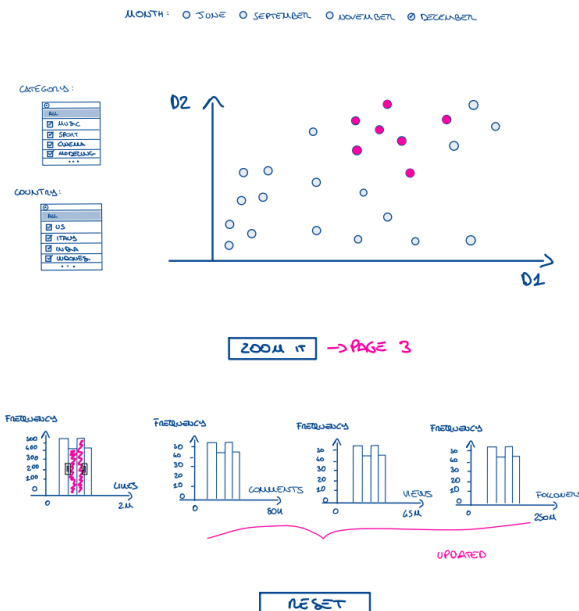
There is then the possibility to "zoom" into a

selection: the page will reload having only the selected youtubers as base dataset, in order to make it easier for him to find what he looks for. Also, the user can at anytime click on a youtuber (or select it) to have details on him. Lastly, when the zoom contains less or equal than 5 youtubers, the user can compare them in a specific view and have insights on them on the 4 months under analysis.

D3.js was used as the main technology behind the solution

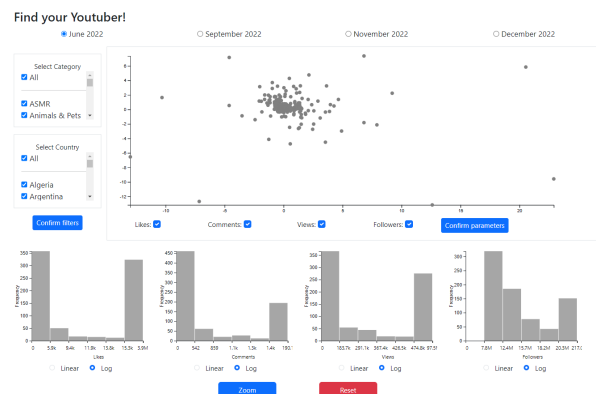
2.2 Design

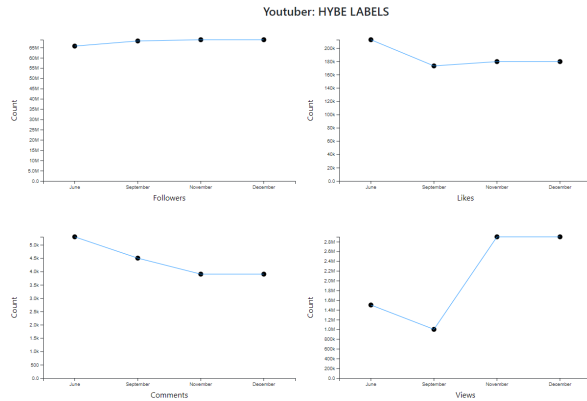
Here are some of the mockups:



2.3 Implementation

The actual software has been developed using D3.js (which makes use of SVG HTML5, and CSS standards) and Flask in order to handle the dimensionality reduction on the dataset.





3 Discovered insights

Through the application of the visual analytics solution, several key insights have been uncovered:

- **Low value channels:** There are many channels with disproportionately high numbers of followers but low levels of engagement, such as T-Series and Sony SAB (the last one having 80M subscribers and an average of 3K views).
- **Category Analysis:** The most performing categories are daily vlog, music and dance, animation, film, and videogames.
- **Clustering:** The majority of YouTube channels (85%) exhibit similar levels of activity
- **Country Trends:** The most popular channels are predominantly from India and America, while countries like Mexico, Pakistan, Philippines, Thailand, India, and Egypt have fewer famous YouTubers.
- **Growth Trends:** YouTube channels tend to grow in number over time.

- **Content Impact:** Certain events, such as the release of a popular song by Shakira in November, can significantly impact channel activity.

4 Related work

To the best of authors knowledge there aren't any visual analytics solution to find specific youtubers for sponsorship or ads.

The proposed related work do not even use infovis techniques. There are indeed some services (which don't use visual analytics) that allow the companies to find specific youtubers for sponsorship or ads, here there will be a comparison on the technical aspect of those works.

4.1 ytubetool.com

This website offers one page with a searchbar for the channel name, a country selector, a filter to choose what to look for (1 option only) between date, rating, view count, relevance, video count. There is no information on what database is being used. The channel name can be left empty to have a wider search. Even though the tool has more filtering options, it doesn't provide any metric on the actual youtuber (there is only a link that redirects to the channel), therefore the only use that one company could do of this software is to leave the channel name empty and have some youtubers ranked, but then check manually one by one.

4.2 channelcrawler.com

The standard package offers a basic search based on country, category, min-max subscribers, min-

max views, min-max number of videos, min-max creation date. It also mentions that it has access to a 20M channels db.

With premium package there are more options, such as if the channel does shorts, is active, has other social medias (specifying which social media) and has specific keywords.

While the database is larger and also the number of filtering options is bigger, the interactability is less since, as the above service, once the data is entered and the filtering is done, there is only a redirect to the channel; no more data is shown and there is no option to tweak the filtering that has been done.

4.3 hypeauditor.com

The website is made of a searchbar that looks for a keyword and finds youtuber related to that. The result is displayed showing the link, the amount of subscribers, the category and an account quality score (established by the website based on audience quality, engagement rate and its authenticity).

This is probably the best service so far, it is the one that goes closest to solving the problem, but it has no more functions than the mentioned ones, therefore leaving not much space for interaction.

5 Conclusion

This solution seems to be the most versatile among all of the previously seen. Even though some of them have some extra feature, they are far from having a visual analytics approach, increasing the average time to get a satisfying result and reducing the interactability.